PROJECT 3

Project Summary

Bank Customer Churn Prediction and KPI Dashboard

Objective

The project uses customer data to understand bank churn patterns, build models that estimate churn risk, and communicate findings through interactive Power BI and Tableau dashboards. The intended audience is a bank's customer-retention or business analytics team.

Dataset and Preparation

The Bank Customer Churn dataset contains 10,000 records and 13 columns. The target variable is Exited. Initial checks found no missing values or duplicate rows. CustomerId and Surname were excluded from the model features to avoid using identifiers as predictive inputs.

Main Findings

Area	Finding
Overall	20.37% of customers in the dataset had exited.
Geography	Germany showed the highest churn rate at 32.44%.
Gender	Female customers had a churn rate of 25.07%, compared with 16.46% for male customers.
Activity	Inactive members had a churn rate of 26.85%, compared with 14.27% for active members.
Products	Churn rates were high in the three- and four-product groups, but those groups contained relatively few customers.

Model Outcome

Three approaches were compared: Logistic Regression, a standard Decision Tree, and a class-weighted (balanced) Decision Tree. The balanced Decision Tree achieved 76.66% recall, 46.15% precision, 57.62% F1 and 84.03% ROC-AUC at the evaluated default classification setting. It was selected for full-dataset risk scoring because it identified a larger share of churners, with the trade-off of more false positives. The dashboard scoring used a probability threshold of 0.60.

Dashboard Outcome

- Power BI presents overall customer and churn KPIs, churn patterns across customer groups, risk-level distribution and high-risk customers by geography.
- A what-if retention-offer calculation estimates acceptances and retained customers using a user-selected acceptance rate and an assumed 40% success rate.
- Tableau Public reproduces key churn views, including overall churn, geography, gender, active membership and risk distribution.

Conclusion

The analysis identifies customer groups that may warrant further investigation and demonstrates how machine learning can support churn-risk prioritisation. The dashboards make the findings easier to explore. The model and scenario estimates should be validated with current data and real campaign experiments before being used for

operational decisions.

Source: Maven Analytics – Bank Customer Churn: https://mavenanalytics.io/data-playground/bank-customer-churn. Add the actual dataset download date and confirm provider usage terms before submission.